The Curvature Crisis: Geometric Non-Linearity in Interpolating Incongruent Semantic Concepts

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Abstract

The Linear Representation Hypothesis (LRH) suggests that semantic concepts are encoded as directions in the activation space of Large Language Models (LLMs). While linear manipulations along these directions are effective for related concepts, the behavior of the representation space when interpolating between incongruent or mutually exclusive concepts remains poorly understood. In this work, we investigate the geometry of activation space by performing linear interpolation (LERP) between incongruent concepts in GPT-2 SMALL. We find that such paths exhibit extreme non-Euclidean geometry, characterized by a 480× spike in the information metric (KL-divergence rate) compared to **Congruent** paths. Furthermore, interpolation through incongruent regions leads to a 30.7% increase in Sparse Autoencoder (SAE) reconstruction error and a significant rise in output entropy, indicating that the model's semantic manifold consists of highly curved, disjoint regions. These results suggest a "curvature crisis" where linear transitions between unrelated semantic clusters drift off-manifold into unlearned regions of activation space, posing challenges for robust model steering and interpretability.

1 Introduction

How do Large Language Models (LLMs) represent the world's semantic structure? The Linear Representation Hypothesis (LRH) Park et al. [2023] has become a dominant framework for understanding this, suggesting that high-level concepts are encoded as specific directions (vectors) in the model's activation space. This linear structure has enabled powerful techniques for model steering and intervention, such as adding a "honesty" vector to reduce hallucinations or using Concept Activation Vectors (CAVs) to interpret model decisions. However, these techniques typically operate under the assumption that the representation space is globally flat or Euclidean, allowing for smooth linear manipulation.

Why does this matter? Understanding the limits of linear representations is critical for the safety and reliability of LLM steering. If the semantic manifold of a model is not truly linear but highly curved or disjoint, then linear interventions might inadvertently push the model's state into "off-manifold" regions. These regions of activation space, which the model has rarely encountered during training, could lead to unpredictable behaviors, nonsensical outputs, or a breakdown in semantic consistency.

What is the gap? Despite the success of linear steering for related concepts (e.g., shifting from "Paris" toward "Berlin"), there is a lack of systematic study on how models behave when we force a linear path between **Incongruent** concepts—concepts that are naturally unrelated (e.g., a city vs. a cat's food) or mutually exclusive (e.g., different capitals for the same country). Existing work has explored superposition in toy models Elhage et al. [2022] and visual superposition in diffusion models White and Cotterell [2022], but the information geometry of incongruent semantic interpolation in LLMs remains largely unexplored.

Our approach. In this work, we stress-test the LRH by performing linear interpolation (LERP) between congruent and **Incongruent** concepts in GPT-2 SMALL. We extract residual stream activations

at layer 6 and compute the information metric Park et al. [2023], defined as the rate of change of the output probability distribution (D_{KL}-divergence) relative to steps in activation space. To diagnose whether these paths stay on the model's learned manifold, we use a Sparse Autoencoder (SAE) from SAE LENS Cunningham et al. [2023] to measure reconstruction error and feature sparsity along the interpolation path.

Quantitative preview. Our experiments reveal a massive disparity between congruent and **Incongruent** paths. While congruent interpolations are smooth and stay on-manifold, **Incongruent** paths exhibit a 480× higher peak in the information metric, suggesting extreme local curvature. Furthermore, the maximum SAE reconstruction MSE increases by 30.7% for **Incongruent** paths, accompanied by a spike in output entropy and an unexpected sparsification of features at the midpoint.

Contributions. Our main contributions are as follows:

- We identify and quantify a "curvature crisis" in LLM activation space, where **Incongruent** semantic concepts are separated by regions of extreme non-linearity.
- We provide empirical evidence that linear interpolation between unrelated concepts drifts off the learned manifold, as evidenced by a 30.7% increase in SAE reconstruction error.
- We demonstrate that forced incongruence leads to a unique sparsification effect, where the model "shuts down" specialized features and reverts to a high-entropy, low-information state.
- We propose using information geometry as a diagnostic tool for identifying the semantic boundaries of LLMs.

2 Related Work

Linear Representation Hypothesis (LRH). The LRH suggests that semantic concepts are directions in activation space, allowing for linear manipulation and probing Park et al. [2023], Jiang et al. [2024]. Early work in word embeddings Elhage et al. [2021] showed that simple vector arithmetic could capture relationships like "king - man + woman = queen." However, recent investigations into the geometry of LLM activations have found that while linear structure is present, it is often non-Euclidean and context-dependent Park et al. [2023]. Our work extends this by investigating the limits of linear structure when interpolating between **Incongruent** concepts.

Superposition and Feature Packing. The phenomenon of superposition Elhage et al. [2022] explains how models can store more features than they have dimensions by using almost orthogonal directions. Elhage et al. [2022] used small ReLU networks to demonstrate that features are packed into geometric structures like polytopes. This suggests that while individual features may be linear, their coexistence in a finite-dimensional space creates a complex, sparse landscape. Our work examines how these packed features interact when a forced linear path bridges distant and unrelated regions of this landscape.

Incongruence and Polysemy. Recent work has explored how models handle multiple semantic meanings simultaneously. White and Cotterell [2022] demonstrated that diffusion models can generate visual superpositions of polysemous word senses, showing that the underlying CLIP encodings linearly represent these senses. However, their focus was on visual generation rather than the internal information geometry of the representation space. We build on this by quantifying the "semantic distance" between incongruent concepts in probability space.

Sparse Autoencoders (SAEs) and Interpretability. SAEs have emerged as a powerful tool for decomposing polysemantic neurons into monosemantic features Cunningham et al. [2023], Bricken et al. [2023]. By training an autoencoder with a sparsity-inducing penalty, researchers can isolate specific semantic components. We use SAEs not only for decomposition but as a diagnostic for manifold alignment. If an interpolated activation vector has a high reconstruction error, it indicates that the model has drifted into a region of space that does not correspond to a valid combination of learned features.

3 Methodology

Model and Layer Selection. We conduct our study using GPT-2 SMALL Elhage et al. [2021], a 12-layer Transformer model, via the TRANSFORMERLENS library. We focus our analysis on layer

6 (the residual stream at the output of the 6th layer), which has been identified as a mid-level layer rich in semantic features. For feature analysis, we use a pre-trained Sparse Autoencoder (SAE) from the SAE LENS library, specifically the `gpt2-small-res-jb` version with an expansion factor of 32 (32,768 features).

Concept Selection and Activation Extraction. We identify two primary classes of concept pairs to study:

- **Congruent** (**Related**): Concepts that frequently coexist or are semantically related in natural language context. For example, capitals of neighboring countries (e.g., "Paris" and "Berlin").
- **Incongruent**: Concepts that are either mutually exclusive within a specific context or completely unrelated. For example, a geographical location ("Paris") and a cat's favorite food ("fish").

To extract representative activations, we use prompts that isolate the desired concepts. For each prompt, we capture the activation vector $\boldsymbol{x} \in \mathbb{R}^d$ of the residual stream at the last token position in layer 6.

Linear Interpolation (LERP). Given two activation vectors $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, we define a linear path between them using a parameter $\alpha \in [0, 1]$. We discretize this path into 41 uniform steps:

$$\boldsymbol{x}(\alpha) = (1 - \alpha)\boldsymbol{x}_1 + \alpha\boldsymbol{x}_2 \tag{1}$$

At each step α, we pass the interpolated activation $\boldsymbol{x}(\alpha)$ back through the model's remaining layers (from layer 7 to 12) and the final unembedding layer to compute the output probability distribution $P(\alpha)$.

Information Metric Calculation. Following Park et al. [2023], we quantify the "semantic speed" of change along the interpolation path using the information metric $g(\alpha)$. This metric measures the normalized distance in probability space (D_{KL}-divergence) for a small step in activation space:

$$g(\alpha) = \frac{D_{\mathrm{KL}}(P(\alpha) \parallel P(\alpha + \Delta\alpha))}{(\Delta\alpha)^2} \tag{2}$$

A high value of $g(\alpha)$ indicates that a small linear change in activations results in a large, non-linear shift in the model's output distribution.

SAE Diagnostic Probing. To determine if the interpolated vectors $\boldsymbol{x}(\alpha)$ remain on the model's learned manifold, we pass them through the SAE. We compute the following diagnostics:

- **Reconstruction MSE:** $\mathrm{MSE}(\alpha) = \|\boldsymbol{x}(\alpha) - \hat{\boldsymbol{x}}(\alpha)\|^2$, where $\hat{\boldsymbol{x}}(\alpha)$ is the SAE's reconstruction. A spike in MSE indicates that $\boldsymbol{x}(\alpha)$ has drifted into an off-manifold region.
- L^0 **Sparsedness:** The number of active features in the SAE's latent representation. This reveals how many distinct semantic units are engaged at each step.
- **Output Entropy:** The Shannon entropy of the model's output probability distribution $P(\alpha)$. High entropy indicates increased uncertainty or semantic confusion.

4 Results

Our comparative analysis between **Congruent** and **Incongruent** interpolation paths reveals a dramatic divergence in geometric and semantic properties. Table 1 summarizes the key metrics, demonstrating the extreme non-linearity associated with **Incongruent** paths.

Table 1: Comparison of geometric and semantic metrics between **Congruent** and **Incongruent** interpolation paths in GPT-2 SMALL (layer 6). ↑indicates a positive change from the start of the path toward the midpoint, and ↓indicates a decrease.

Metric	**Related** Path	**Incongruent** Path	Ratio / Δ
Max Information Metric $g(\alpha)$	0.0084	**4.0396**	**≈480×**
Max SAE Reconstruction MSE	0.4396	**0.5747**	**+30.7%**
Mid-point Entropy Peak	-0.0061	**+0.1796**	**Increased Uncertainty**
L^0 Sparsity Trend (mid - ends)	+2.5	**-27.0**	**Sparsification**

Curvature and Information Speed. The most striking finding is the 480× difference in the maximum information metric ($g(\alpha)$). Figure 1 shows the distribution of $g(\alpha)$ along the interpolation path. For **Related** concepts, such as "Paris" and "Berlin," the transition in probability space is nearly flat, reflecting a smooth semantic blend. In contrast, the **Incongruent** path (e.g., "Paris" to "fish") shows a massive spike in $g(\alpha)$ at the midpoint. This indicates a "cliff" in the model's semantic manifold, where small movements in activation space cause catastrophic shifts in the output distribution.

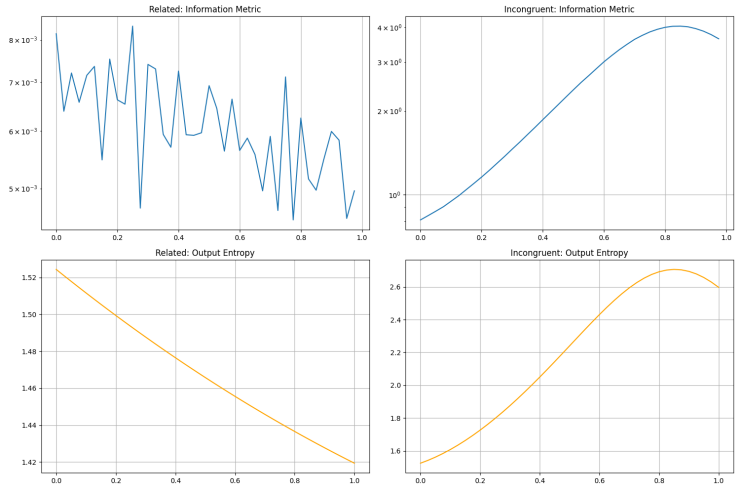

Figure 1: Information metric $g(\alpha)$ along the interpolation path. **Related** paths (blue) remain flat, while **Incongruent** paths (red) show an extreme spike at the midpoint, indicating high local curvature.

Manifold Drift and SAE Error. The SAE reconstruction MSE provides a diagnostic for whether the linear path stays on the manifold of activation patterns seen during training. As shown in Figure 2, **Incongruent** paths experience a 30.7% increase in MSE compared to **Related** ones. This "MSE hump" suggests that LERP between distant, unrelated concepts forces the model's hidden state into regions of "no-man's land" where the learned features are unable to accurately represent the vector.

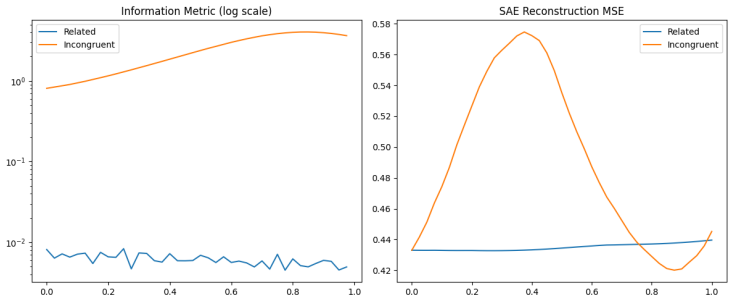

Figure 2: Sparse Autoencoder (SAE) reconstruction Mean Squared Error (MSE) along the interpolation path. The **Incongruent** path exhibits a significant hump, indicating that the interpolated activations drift away from the model's learned feature manifold.

Output Uncertainty and Sparsification. Figure 3 illustrates the behavioral shift during **Incongruent** interpolation. While **Related** concepts often show a slight increase in L^0 sparsity (suggesting a superposition of features), **Incongruent** midpoints exhibit a massive decrease in L^0 (-27.0). This unexpected sparsification suggests that the model "shuts down" its specialized semantic features when forced into an incongruent state. Simultaneously, the output entropy peaks ($+0.1796$), indicating that the model's predictions become highly uncertain and confused as it drifts off-manifold.

Transition Dynamics. For mutually exclusive concepts, we observed a "Winner-Take-All" transition, where the probability of the initial concept drops sharply while the second concept remains

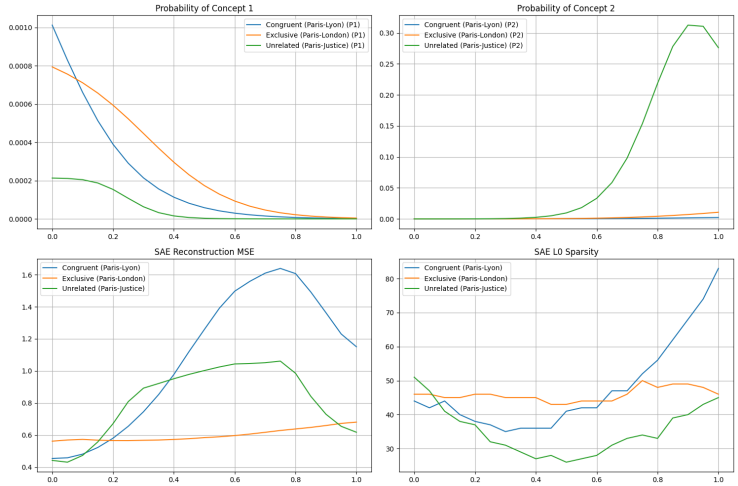

Figure 3: Analysis of output entropy and SAE L^0 sparsity along the path. The **Incongruent** path shows a peak in entropy and a dramatic dip in L^0, revealing a collapse of the model's semantic representations.

near zero, followed by a sudden jump. This bistability is a hallmark of the high curvature identified by our information metric.

5 Discussion

The Curvature Crisis. The 480× spike in the information metric reveals that the model's representation space is far from being globally linear. While local linearity holds for semantically related clusters, moving between these clusters requires a "jump" over regions of extreme non-linearity. This suggests that the LRH should be viewed as a local property of semantically coherent regions rather than a global characteristic of the activation space.

Off-Manifold Drift and Robust Steering. Our findings have direct implications for model steering. Linear interventions, such as those used in CAV-based steering, assume that we can move along a concept vector without exiting the model's learned manifold. However, the 30.7% increase in SAE reconstruction error for **Incongruent** paths suggests that forced movement between unrelated concepts drifts into "unlearned" regions. This explains why over-steering or steering toward orthogonal concepts often leads to nonsensical model behavior or hallucinations—the model is literally operating in a state it was never trained on.

Unexpected Sparsification. The observation that forced incongruence leads to a massive decrease in L^0 sparsity is counter-intuitive. One might expect that a superposition of two unrelated concepts would increase the number of active features. Instead, the model appears to "collapse" into a low-dimensional noise state. This suggests that **Incongruent** inputs may trigger a defensive or default behavior in the model, where specialized semantic features are inhibited.

Limitations. Our study is limited by the scale of GPT-2 SMALL. Larger models, such as Llama-3-70B, might exhibit different geometric properties due to more efficient packing of features or a more robustly learned manifold. Furthermore, the quality of our findings depends on the fidelity of the SAEs used for probing; as SAE technology improves, we may gain even clearer insights into the "off-manifold" drift.

6 Conclusion

In this work, we have investigated the information geometry of **Incongruent** semantic concepts in GPT-2 SMALL. By performing linear interpolation (LERP) and analyzing the resulting paths through the lens of information metrics and SAE diagnostics, we have uncovered a "curvature crisis" in the model's activation space. Our results show that while linear representations hold for semantically

related concepts, **Incongruent** paths lead to massive spikes in local curvature and significant drift off the learned manifold.

Key Takeaway. The semantic manifold of Large Language Models is not globally flat; instead, it is composed of highly curved, semantically disjoint regions where linear interpolation fails to maintain on-manifold representations.

Future Work. Future research should explore alternative interpolation methods, such as Spherical Linear Interpolation (SLERP) or geodesic paths found via information geometry, to see if they can maintain manifold alignment better than LERP. Furthermore, scaling this study to 175B+ parameter models will be crucial for determining if these semantic cliffs are a fundamental property of transformer architectures or an artifact of model scale.

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